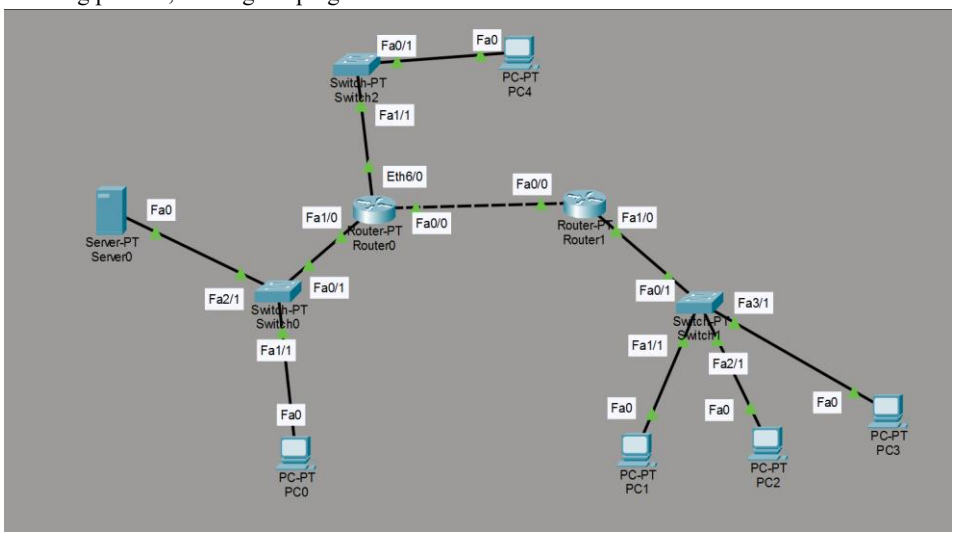
Lab 6

Commented [1]: 7-3

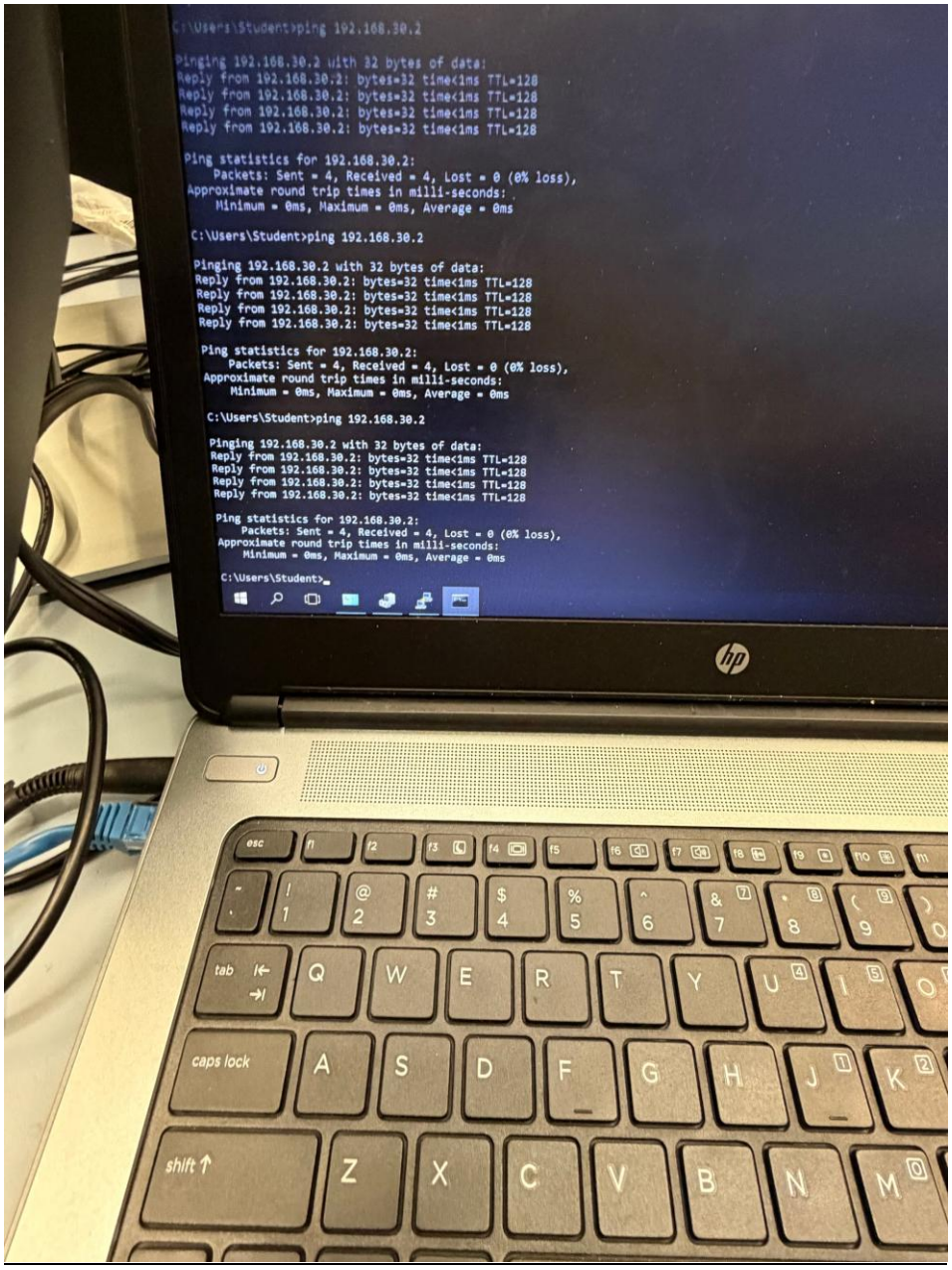
In the physical configuration, everything is the same as the Cisco packet tracer topology. The only difference is that the firewalls were disabled on the PCs in the physical setup due to it blocking packets, causing the ping commands to fail.

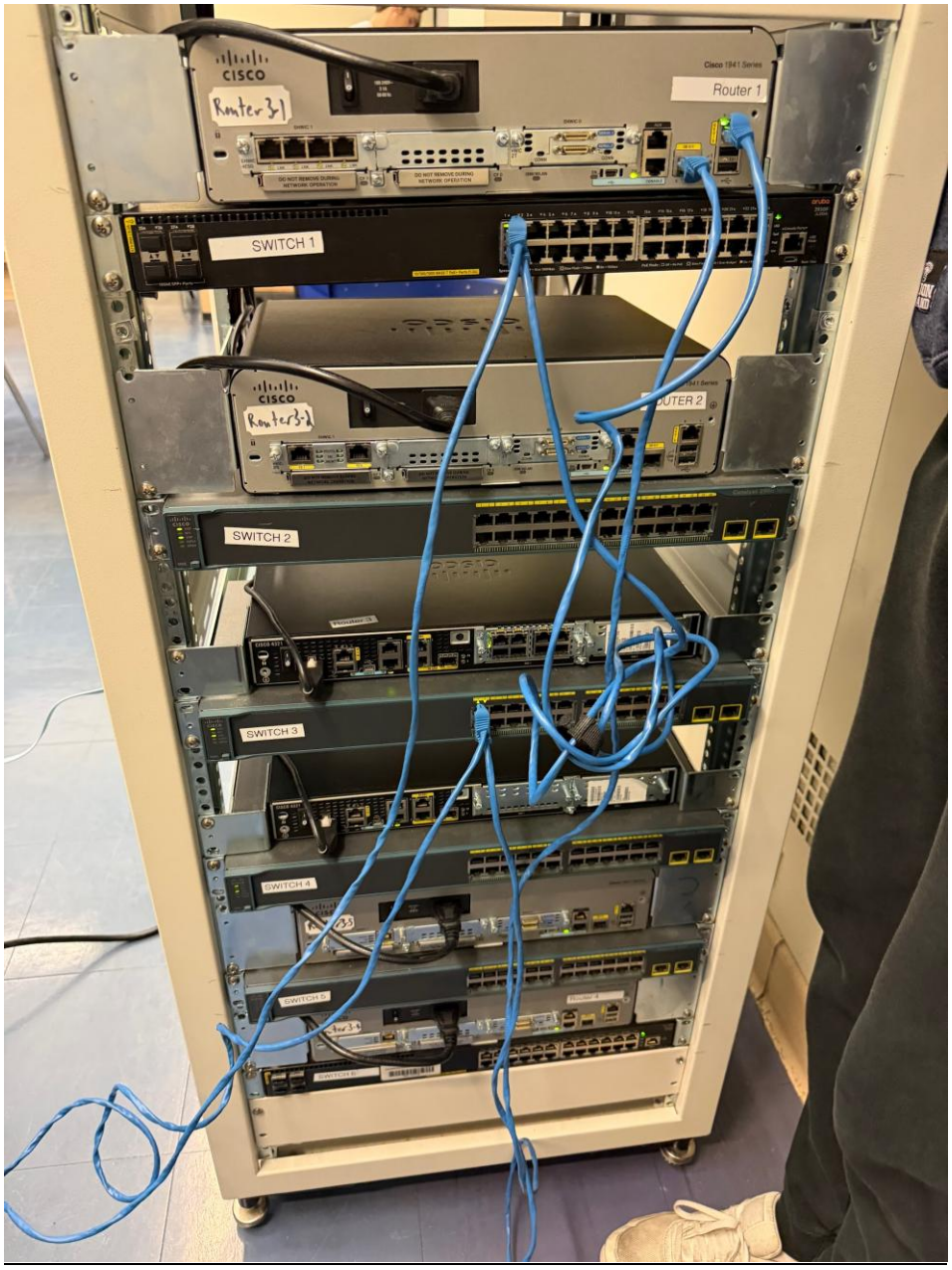


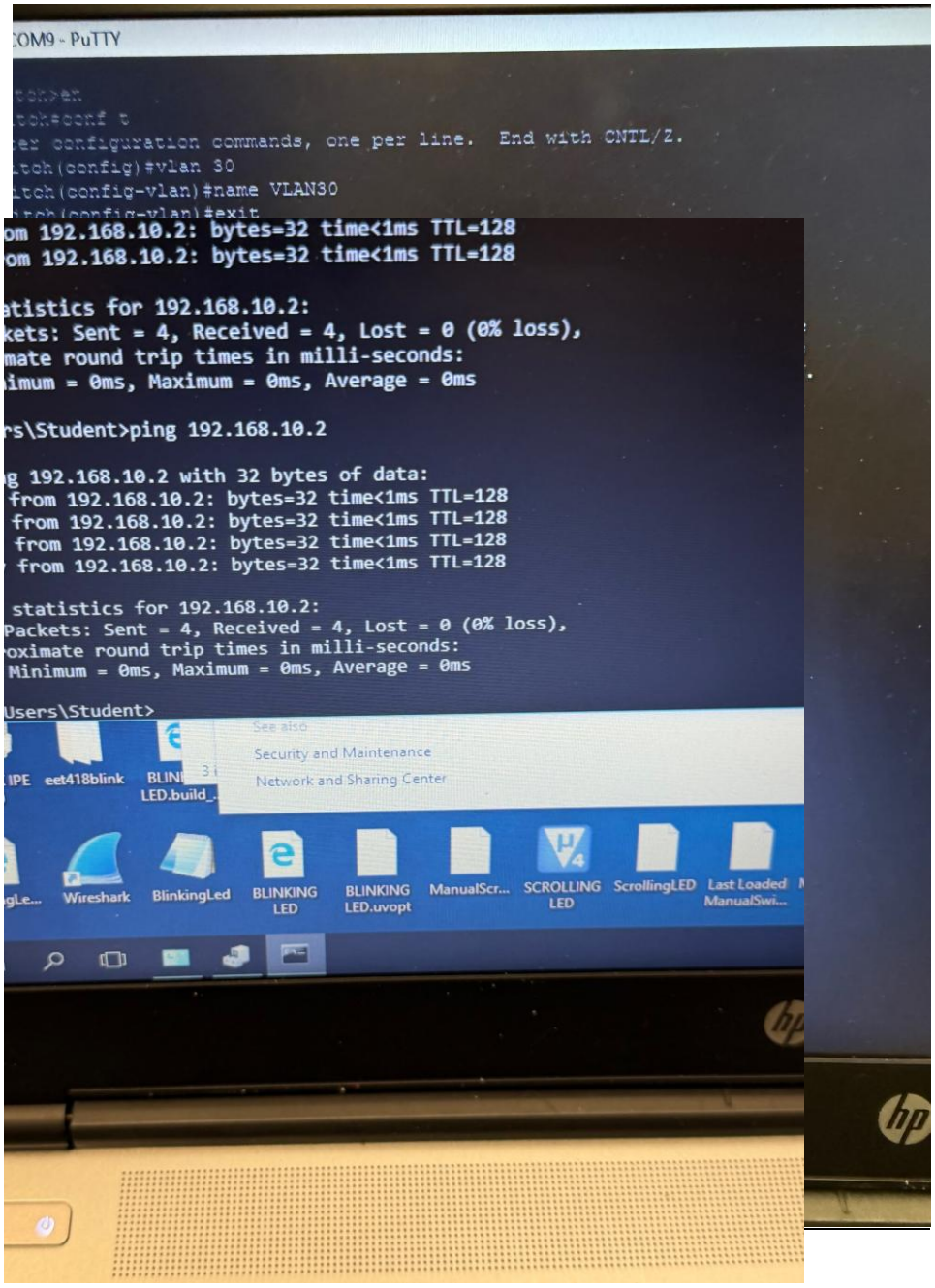
Simulation Result Pkt Tracer:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	Server0	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC4	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC0	PC3	ICMP		0.000	N	3	(edit)	(delete)

Hardware:


C:\Users\Student>ping 192.168.30.2
Pinging 192.168.30.2 with 32 bytes of data:
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.30.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Student>ping 192.168.30.2
Pinging 192.168.30.2 with 32 bytes of data:
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.30.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Student>ping 192.168.30.2
Pinging 192.168.30.2 with 32 bytes of data:
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Reply from 192.168.30.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.30.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Student>_


CISCO
Router 3-1
Cisco 1941 Series
Router 1
SWITCH 1
CISCO
Router 3-2
ROUTER 2
SWITCH 2
SWITCH 3
SWITCH 4
SWITCH 5


COM9 - PuTTY
tch>en
tch#conf t
er configuration commands, one per line. End with CNTL/Z.
itch(config)#vlan 30
itch(config-vlan)#name VLAN30
itch(config-vlan)#exit
om 192.168.10.2: bytes=32 time<1ms TTL=128
om 192.168.10.2: bytes=32 time<1ms TTL=128
atistics for 192.168.10.2:
kets: Sent = 4, Received = 4, Lost = 0 (0% loss),
mate round trip times in milli-seconds:
imum = 0ms, Maximum = 0ms, Average = 0ms
rs\Student>ping 192.168.10.2
g 192.168.10.2 with 32 bytes of data:
from 192.168.10.2: bytes=32 time<1ms TTL=128
from 192.168.10.2: bytes=32 time<1ms TTL=128
from 192.168.10.2: bytes=32 time<1ms TTL=128
from 192.168.10.2: bytes=32 time<1ms TTL=128
statistics for 192.168.10.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
oximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Users\Student>
See also
Security and Maintenance
Network and Sharing Center
IPE
eet418blink
BLINK LED.build_
Wireshark
BlinkingLed
BLINKING LED
BLINKING LED.uvopt
ManualScr...
SCROLLING LED
ScrollingLED
Last Loaded ManualSwi...

CONFIGURATION COMMAND FOR PACKET TRACER:

Router 0:

```
enable
configure terminal
hostname Router0

interface fa1/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit

interface eth6/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit

interface fa0/0
 ip address 192.168.40.1 255.255.255.0
 no shutdown
exit

ip route 192.168.30.0 255.255.255.0 192.168.40.2

end
write memory
```

Router 1:

```
enable
configure terminal
hostname Router1

interface fa1/0
 ip address 192.168.30.1 255.255.255.0
 no shutdown
exit

interface fa0/0
 ip address 192.168.40.2 255.255.255.0
 no shutdown
exit
```

```
ip route 192.168.10.0 255.255.255.0 192.168.40.1
ip route 192.168.20.0 255.255.255.0 192.168.40.1

end
write memory

Switch 0
enable
configure terminal
hostname Switch0

interface fa1/1
 switchport mode access
 no shutdown
exit

interface fa2/1
 switchport mode access
 no shutdown
exit

interface fa1/0
 switchport mode access
 no shutdown
exit

end
write memory

Switch 1
enable
configure terminal
hostname Switch1

interface fa1/1
 switchport mode access
 no shutdown
exit

interface fa2/1
```

```
 switchport mode access
 no shutdown
exit

interface fa3/1
 switchport mode access
 no shutdown
exit

interface fa1/0
 switchport mode access
 no shutdown
exit

end
write memory
```

Switch 2

```
enable
configure terminal
hostname Switch2

interface fa0/1
 switchport mode access
 no shutdown
exit

interface fa1/1
 switchport mode access
 no shutdown
exit

end
write memory
```

CONFIGURATION COMMAND FOR PUTTY:

MLS0 / Main Switch

```
enable
configure terminal
```

```
vlan 10
 name VLAN10
exit

vlan 20
 name VLAN20
exit

vlan 30
 name VLAN30
exit

interface fa0/1
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 no shutdown
exit

interface fa0/2
 switchport mode access
 switchport access vlan 10
 no shutdown
exit

interface fa0/3
 switchport mode access
 switchport access vlan 20
 no shutdown
exit

interface fa0/4
 switchport mode access
 switchport access vlan 30
 no shutdown
exit

end
write memory
```

Second Switch

```
enable
configure terminal

vlan 30
 name VLAN30
exit

interface fa0/1
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 no shutdown
exit

interface fa0/2
 switchport mode access
 switchport access vlan 30
 no shutdown
exit

interface fa0/3
 switchport mode access
 switchport access vlan 30
 no shutdown
exit

end
write memory
```

Verification Commands

```
show vlan brief
show ip interface brief
show running-config
```

CONCLUSION & DISCUSSION:

This lab demonstrated how multiple LANs can be connected through routers and switches so that devices on different networks can communicate successfully. Separate networks were created for the 192.168.10.0/24, 192.168.20.0/24, and 192.168.30.0/24 subnets, and routing was configured between the two routers so that traffic could pass from one LAN to another. After assigning IP addresses, default gateways, and routing commands, the devices were able to successfully ping

hosts on remote networks, which confirmed that the configuration was working correctly. The Packet Tracer simulation helped verify the logical topology, while the PuTTY configuration showed how the physical switches were set up in the lab environment. Overall, this lab reinforced the concepts of IP addressing, switch port configuration, and inter-network communication through static routing.